

A physics-informed matrix engine bypasses scaling relations in alkyne-poisoned catalytic interfaces —

Part 2

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1 Methodology & Mathematical Derivations

1.1 Physical Justification for the Tight-Binding Framework

Traditional electronic descriptors, such as the static d -band center, collapse the entire electronic structure of a catalytic surface into a single scalar value. This approximation holds well for simple, single-atom adsorbates like atomic oxygen or hydrogen [1]. However, it fails fundamentally for polyatomic molecular adsorbates like terminal alkynes ($R-C \equiv H$). Alkyne adsorption is inherently a multi-center phenomenon governed by the Dewar-Chatt-Duncanson (DCD) paradigm [2]. This interaction involves two simultaneous, competing pathways: σ -donation from the alkyne π -orbital to empty metal d -orbitals, and π -backbonding from occupied metal d -orbitals into the alkyne π^* -antibonding orbital.

To capture this multi-center orbital mixing without the massive computational expense of periodic DFT, we select a tight-binding framework based on a modified three-center Hückel-Coulson Hamiltonian. This choice is justified by three key factors:

1. **Computational Efficiency:** It reduces the electronic structure of the interface to its essential frontier orbitals, allowing candidate configurations to be evaluated in less than 0.2 milliseconds.
2. **Physical Interpretability:** By solving the secular equations analytically, we can directly separate individual electronic contributions that are usually lumped together in black-box machine learning models.
3. **Dynamic Response:** Unlike static descriptors, this model accounts for changes in the active-site wavefunction as a function of coordination distance (r) and the local chemical environment.

1.2 Construction of the Localized Tight-Binding Hamiltonian

We model the interface using a localized three-center basis set comprised of the active metal site orbital ($|M\rangle$), the terminal alkyne carbon atom orbital ($|C_1\rangle$), and the internal alkyne carbon atom orbital ($|C_2\rangle$). In this localized basis, the effective real symmetric tridiagonal Hamiltonian matrix

(**H**) is written as:

$$\mathbf{H} = \begin{pmatrix} \alpha_M & \beta_{M-C1}(r) & 0 \\ \beta_{M-C1}(r) & \alpha_{C1} & \beta_{C\equiv C} \\ 0 & \beta_{C\equiv C} & \alpha_{C2} \end{pmatrix} \quad (1)$$

Where α_M represents the effective on-site energy of the active metal site, α_{C1} and α_{C2} represent the on-site energies of the carbon atoms, and $\beta_{C\equiv C}$ is set as a constant structural parameter (-7.20 eV). The interfacial coupling is governed by a modified Wolfsberg-Helmholz approximation combined with an exponential overlap decay function [3]:

$$\beta_{M-C1}(r) = k_{\text{scale}} \cdot \exp[-\kappa(r - r_0)] \cdot \left(\frac{\alpha_M + \alpha_{C1}}{2} \right) \quad (2)$$

Where $k_{\text{scale}} = 1.75$, $\kappa = 1.2 \text{ \AA}^{-1}$, and $r_0 = 1.2 \text{ \AA}$.

1.3 Step-by-Step Derivation of Eigenvalues and Wavefunctions

To determine the electronic energy states, we solve the secular determinant:

$$\det(\mathbf{H} - E\mathbf{I}) = 0 \quad (3)$$

Expanding this determinant along the first row yields the characteristic cubic polynomial equation:

$$(\alpha_M - E) [(\alpha_{C1} - E)(\alpha_{C2} - E) - \beta_{C\equiv C}^2] - \beta_{M-C1}^2(\alpha_{C2} - E) = 0 \quad (4)$$

The lowest root corresponds to the primary bonding ground-state eigenvalue (E_{bond}) [4]:

$$E_{\text{bond}} = \frac{1}{3} \text{Tr}(\mathbf{H}) - 2\sqrt{Q} \cos\left(\frac{\theta}{3}\right) \quad (5)$$

Once the eigenvalues are established, we evaluate the wavefunctions to determine the dynamic quantum descriptors. The local π -backbonding susceptibility (ΔE_π) is isolated via second-order perturbation theory relative to the unmixed alkyne LUMO level ($\alpha_{\text{LUMO}} = \alpha_{C1} - 0.5$ eV):

$$\Delta E_\pi = \frac{\beta_{M-C1}(r)^2}{\alpha_M - \alpha_{\text{LUMO}}} \quad (6)$$

Using Coulson's population analysis method, the net ground-state charge density localized at the terminal carbon atom (C_1) is determined assuming a two-electron filling of the lowest energy channel [5, 6]:

$$q_{C1} = 1.0 - 2(c_{C1,1})^2 \quad (7)$$

Where $c_{C1,1}$ is the exact terminal carbon eigenvector coefficient derived from the system secular components:

$$c_{C1,1} = \frac{\beta_{M-C1}(E_{\text{bond}} - \alpha_{C2})}{\sqrt{\beta_{M-C1}^2(E_{\text{bond}} - \alpha_{C2})^2 + (E_{\text{bond}} - \alpha_M)^2(E_{\text{bond}} - \alpha_{C2})^2 + \beta_{C\equiv C}^2(E_{\text{bond}} - \alpha_M)^2}} \quad (8)$$

References

- [1] Nørskov, J. K. et al. *Nature Chem.* **1**, 22–30 (2009).
- [2] Chatt, J. & Duncanson, L. A. *J. Chem. Soc.* 2939–2947 (1953).
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- [4] Wilkinson, J. H. *The Algebraic Eigenvalue Problem*. (Clarendon Press, 1965).
- [5] Coulson, C. A. *Proc. R. Soc. Lond. A* **169**, 413–428 (1939).
- [6] Murrell, J. N. et al. *The Chemical Bond*. (Wiley, 1978).

Data and Code Availability

The high-throughput screening dataset containing the calculated quantum metrics (ΔE_π and Δq_{C1}) for the 300,000 catalyst–reactant configurations, along with the foundational periodic DFT parameters, has been permanently archived on Zenodo under DOI: 10.5281/zenodo.21196337. The complete, self-contained Python script used to verify these analytical metrics and drive the screening matrix engine is provided below: